

Machine Learning Lecture 3 Summary – Group 720

Download full dataset, which is a Matlab format of [THE MNIST DATABASE of handwritten digits](#): A training set of 60,000 examples, and a test set of 10,000 examples; each having a size of $28 \times 28 = 784$ (concatenating the pixels). Do exercise:

(1) from the 10-class database, choose three classes (5, 6 and 8) and then reduce dimension to 2 using LDA

(Code implemented in the python file)

(2) perform 3-class classification based on the generated 2-dimensional data.

We obtained the following classification accuracy using LDA: 94.48%

This means the model can classify the digits very good despite there only being 2-dimensions.